

èfjç» - ç” µè`❖i¼š188-5665-3017 - é,®ç®±i¼šhs3434@foxmail.com -
ä, ºäºç½ ‘ç«™i¼šhs3434.github.io - **æ±
,è❖Ææ,,❖â❖ ‘**i¼šAMD ROCm Python â❖Žç«`â¼â❖ ‘â • ¥ç`â, ^i¼^ROCm Radeon Cloud
æ-¹â❖ ‘i¼%º --- ## ä, ºäºç®€ä» 3 â¹´ç” Ýç%ºâžjæ❖`â • ¥ç`âÆ-ç»❖éâEi¼Æè ± ââè½-âž`
K8s â❖Žç«`æ ^â€, \*\*â®¹â™™âÆ-éƒ`ç½²â€❖èµ,,æº❖èºƒâº|â€❖ä»»âŠjç¼-æž’ â€❖GPU
æž`ç❖ + æœ❖âŠj\*\*â€, æœ€èž ‘ 1 â ‘`è ± ââ| K8s / Helm i¼Æç«-ç«`â®Ææ^❖ [**k8s-llm-
runtime**](https://github.com/hs3434/k8s-llm-runtime)i¼^K8s ä, Šçš,, vLLM
æ`jâž`æœ❖âŠjç½ ‘â...³i¼%ºi¼Æ**v0.1.0 tagged i¼Æâ❖`çž`âœºè • ‘demo**â€, ç + Ýæ,%º
Snakemake ç±» Workflows çš,, ML â¹³â❖ºæš½è±jâ€, ## æš€æœ`â❖šâ®ç - [Transformer
â»žâ...¥è—`â^ºä, Šæ%º](https://hs3434.github.io/2025/03/03/transformer1/) - [GPT éžžæ ¼
Transformer è Š £ç ❖ EEGi¼šä, ºä»æ¹^â€❖æ€žâ¹^â❖šâ€❖è, ©âº + ä»æ¹^â❖ ‘](https://
hs3434.github.io/2025/06/08/transformer-eeg-decoding/) - [â»žžé¶ æ❖ä, €â¥—ç«`â^ºç«` EEG
âžjâ❖ • â¶,ç❖ + â • ¥â... • é “¾](https://hs3434.github.io/2025/06/08/eeg-signal-processing-
toolchain/) --- ## æ±,è❖Ææš€èƒ½i¼^æÆ%ºâ²—â½❖âÆ¹é...❖æž’ âº❖i¼%º | ç±»â^« |
æš€èƒ½ | |-----|-----| | **â❖Žç«`æ , âžj** | Python 3.11+â€❖FastAPIâ€❖pytest | |
**â®¹â™™`ä, Žç¼-æž’ ** | Dockerâ€❖Kubernetesâ€❖Helmâ€❖é • œâƒâ❖ä» “âº “çºjç❖ + | |
GPU / LLM ç” Ýæ€❖ | vLLMâ€❖PyTorchâ€❖Hugging Face | | **â • ¥â... • é “¾** |
uvâ€❖mypyâ€❖GitHub Actions | | **èžç»`\*\* | Linux èžç»` | | **æ • ºæ❖âº “** |
PostgreSQLâ€❖MySQLâ€❖Redis | --- ## èjçºç»❖éâE ### k8s-llm-runtimei¼šK8s ä, Šçš,, vLLM
æ`jâž`æœ❖âŠjç½ ‘â...³ - è ± ââ|demoéjçº **è—®éç`**i¼^â¹âº” JD è❖Æè´£ 1 +
3i¼%ºi¼švLLM æ`jâž`éƒ`ç½²éœ GPU + â®¹â™™ + è • `ç” ±i¼Æâºçé`Ýâ¶šâºâ...±âºæ—
¶ æ`jâž`çºjçº + âºæ❖, i¼ç” çº ¶ /âºâºçé`Ýæƒ³ç” ” OpenAI â...¼â®¹ API æ— ¶ é—
” æ Š ´æ«`â€, i¼^çº æžžâ¹âº” JD i¼š”â¶ Š æ`jâž`èç»ƒ/æž`ç❖ +
Workshopâ€❖æ • ™ç`â€❖âšžèƒ½æ¼” çºº”i¼%º **æž ¶ æž,,**i¼^â❖Æ chart + ä, %ºâ±, lib +
é • œâƒâ¼^âÆ-i¼%ºi¼š - `charts/llm-inference/`i¼švLLM è`Ýè½½ charti¼Æ**GPU vendor
â^ ± æ❖çä»...é❖ `values.yaml` ** i¼^amd.com/gpu / nvidia.com/gpu / none
ä, %ºæ€i¼Ævalues æ`¾â¼â❖é€%ºæ©i¼%º - `charts/llm-router/`i¼šFastAPI ç½ ‘â...³ chart -
RBAC æœ€âº❖æ❖ƒé™❖i¼^Pod/Service/ConfigMap read+write + Leasei¼%º - `src/
k8s_llm_runtime/`i¼š3 â±, lib æš½è±j - `K8sJobOperator`i¼^â½žâ±, i¼Æéšç” ” Job
èºƒâº|i¼%º - `VLLMInferenceOperator`i¼^â, â±, i¼Æhelm CLI subprocess + GPU vendori¼%º -
`ModelOperator`i¼^é«`â±, i¼ÆOpenAI â...¼â®¹ + è ± âšš`éƒ`ç½²i¼%º **4
â¶ Š æš€æœ`âºç, **i¼š 1. **lib ä, %ºâ±, â^ + â±, æš½è±j**i¼^â¹âº” JD
â❖Žç«`æž ¶ æž,,èƒ½âšž i¼%º - ä, Šâ±, ModelOperator â❖ââ...³âžƒ load_model(alias) /
unload_model(alias) - ä, â±, VLLMInferenceOperator âºè£... helm ç» + èš, - â½žâ±
, K8sJobOperator éšç” ” K8s Job èºƒâº| 2. **Helm ConfigMap æ³`â...¥ aliases**i¼^â¹âº” JD
â®¹â™™âÆ- + èµ,,æº❖èºƒâº|i¼%º - é❖žâ...❖æ`æ-æ-ºâžžæ`jâž`éƒ½ rebuild
é • œâƒâi¼^GPU é • œâƒâ 15-20GB rebuild 5-10mini¼%º - ConfigMap
ç Š ´ çº Š ç” Ýæ • ^i¼Æé¶ ¶ é • œâƒâé ± âºº 3. **K8s Lease â + â, ƒâ¼â❖é” ❖**i¼^â¹âº”
JD é + ç¾èžç»` + HAi¼%º - coordination.k8s.io/v1 Lease é²æç Router
â¶šâºæœ-â¹ ¶ â❖ ‘éƒ`ç½²â❖Æä, €æ`jâž`- é” ❖ç²’ âºæÆ%º model
aliasi¼Æè ± âšš`èž ± æœÝ + ç»çº| 4. **ä, šâŠjçº Š Prometheus metrics**i¼^â¹âº” JD
â¹³â❖â❖`è Š, æµæ€ Š i¼%º - `chat_completions_duration_seconds`i¼^histogrami¼Æâ❖«
status code æ ± ç¾i¼%º - `http\_requests\_total`i¼^counteri¼ÆæÆ%º method/route/statusi¼%º
- `models\_loaded`i¼^gaugei¼Æâ❖æ~ in-memory statei¼%º **GPU vendor â^ ± æ❖ç
demo**i¼^â¹âº” JD âšš â^ + GPU é + ç¾èi¼%ºi¼š - values.yaml `gpu.vendor=amd|nvidia|
none` - AMDi¼šamd.com/gpu èµ,,æº❖èºƒâº|i¼ÆROCm vLLM é • œâƒâ - NVIDIAi¼šnvidia.com/
gpu èµ,,æº❖èºƒâº|i¼ÆCUDA vLLM é • œâƒâ - nonei¼šCPU only æ`jâ¼âi¼^kind cluster

é» è"ëi¼Ææœ-âœ°â¼Æâ" ç" "i¼%00 **ë" ' 'ë ÷ ðä¿ë-ð"i¼š - 31 commits / 90 tests pass / coverage 88% - ruff + mypy strict + helm-lint â... " clean - 2 Helm chartsi¼^llm-inference + llm-routeri¼%00 **ë, ©â" 'ä, Žä¿âð"i¼š - kind èš,ç, 'â¼â ÷ ...

é • œâfðâš éëYi¼šcontainerdConfigPatches + DaoCloud mirror ### âY°äŽ Snakemake çš,, ML â¹³âð°â • ¥ä½œæµðæj ÷ æž ¶i¼^ç" Yä¿jèfÆæ™~i¼%00 - èfÆæ™~i¼šSnakemake rule è - æ³ • æ— æ³ • âžžçŽœ"jâð—ésðæ,,ðç»,âð - è³¼è®j¼šæ "â¼f rule è - æ³ • i¼Æç" 'æž¥è°fâ ÷ ...éf"æž¥âð£âš"æëðç" Yæ^ð rulei¼Æè š,,èÆfè¾ "ä... ¥è¾ "â ÷ ç±»âž:âð³âð-ç š - æœ"â¼ðæç¼æž¥ - â...³èð" i¼šJD âš â^ ÷ éj"æœ°â™"â - |â¹â¹³âð°ç" âð "ç»ðéâÆ"âë" âë" æœ-éj¼ç"æ"~ ML â¹³âð°æš½è±jçš,,â • ¥ç" 'âžè • µ - **è-|ç» ÷ âðšâðç"i¼š[ä, ðæ»j Snakemake çš,,éð™æëð rulei¼Yæ^ "ç" " Python â¹è± jâš"æëðæž,,â°â ÷ ä, ëæðj RNA-seq æµðæ" 'ç°¿](https://hs3434.github.io/2025/06/08/rnaseq-pipeline-engineering/) ### è,, 'æœ°æž¥âð£ CNN è š £ç ðâ™"i¼^PyTorch¼%00âë" âë" è ÷ â¼çš,,demoéj¼ç" - çç - â°çç - æ— ¶ ç°âð • ç š - è š £ç ð EEG ä¿jâð • - PyQt âžæ - ¶ GUI + FastAPI æš 'éœ² - **ä» "â" ***i¼šgithub.com/hs3434/bci-pipeline-demo --- ## â • ¥ä½œç»ðâž ÷ ### ä, šæµ • â°

æë ÷ Æâðjâ¿jæð-ç š "æš£æœ°00é™ðâ...-âð, i½œç" Yç%00â¿jæð-â • ¥ç" 'â, ^ i½œ 2025.04 - è ÷ ³äš - **rnaseq**i¼^éj¼ç"â, »âš¼i¼%00i¼šâY°äŽ Snakemake çš,, RNA-seq è ÷ âš"âÆ-â^ ÷ æžð pipelinei¼ÆDocker / Apptainer â°¹â™"âÆ-i¼Æ20 æ, â¹ ¶ èjÆi¼»config.yaml é©±âš"é...ðç½°i¼»â...-âð, ä, šâšjè ÷ âš"âÆ-ä, »ç°¿ - **helix**i¼^Python âðžçç - APIi¼%00i¼šFastAPI + Click + Pydantic v2

âðžçççš,,ç" Yç%00â¿jæð- API æœðâšj¼Æç»™ rnaseq pipeline æððä¾ç» "æžœâ± • çð°â±,i¼»ç±»âž:âÆ- ÷ æµè - èj ÷ ç--

æœðâšjçç - è¿ðçç" *i¼šLinux æœðâšjâ™" æ-¥â, ç» 'æšð¼ÆNginx âððâð "ä»£çð ÷ i¼Æsystemd æœðâšjç°jçð ÷ ### ä, šæµ • æ-š æ" "ç" Yç%00âÆ»â - |ç š "æš£æœ°00é™ðâ...-âð, i½œç" Yç%00â¿jæð-ç" âð "â • ¥ç" 'â, ^ i½œ 2022.08 - 2024.07 - **ä" 'â¹³âð°ç" Yä¿jâ • ¥â... • â¼Æâð " **i¼šR / Python / Linux / Docker / â" âžYç" Y - **ä, šâšjè ÷ âš"âÆ- **i¼šâÆ»â|è½-â½ • ç»,ä, šâšjæµðç" 'è ÷ âš"âÆ-i¼Ææ"jâð—âÆ-âëðâð-âðçç" " - **â¹³âð°âœ°âðç"i¼š[cloud.oebiotech.com](https://cloud.oebiotech.com/#/home) ### ä, šæµ • æ-š æ" "ç" Yç%00âÆ»â|æœ°00é™ðâ...-âð, i½œç" Yç%00â¿jæð-â°žâ¹ç" Y i½œ 2021.10 - 2022.01 - âð • ç» ÷ èfžè½-â½ • ç»,i¼šè "æž š âëðâ, è š,,â^ ÷ æžðæëðâ" ®ðžâëðä, æë š âÆ-â^ ÷ æžð - ä, »èjðæš£æœ-i¼šâð • ç» ÷ èfžè½-â½ • ç»,âëðRâëðPythonæëðLinux --- ## æ • ™è,²èfÆæ™- **è¥¿âÆ-â ÷ œæž-ç š "æš£âð š âj"i½œ æðç%00ç š "â - |ä, Žæš£æœ-ä, "ä, š i½œ æœ-ç š "i½œ 2018.09 - 2022.07 i½œ GPAi¼š3.35 - æ" • ä, šè°æ- ÷ i¼šâëšâY°äŽéë»è¾ "æ-è" ,âžâ½" ç¼°â°šâ°é°|âë' ä¼ðç»,â...³æë š çš ¶ âðšâš-âë' æë š çš ¶ æÆ ÷ æ ÷ âëç - âð š â^éj¼ç"i¼šæ¹ç~èðÆæž¥ç š ðâ¹ç" «èš±èœè "¿æš— é "œæ±jæY "çš,,â½±â "ð"i¼^âð,ä, Žä½œèë...i¼%00 --- ## ä, â°°â¼~âš¿ - **1 â "è ÷ â¼ K8s âðžçç - æ ^**i¼šâ»žé ¶ â^° v0.1.0 tagged éj¼ç"i¼Æçç-çç«âÆæ^ðâðÆ Helm chart + Python lib - **çž°âœ° demo èf½âš"i¼šâð-è ÷ ââ, |ç" µè,, 'âœ" kind cluster ä, šè • 'éëšâÆæ • ' LLM æžçð ÷ æµðç" i¼^éf"ç½² â ÷ "è - ç" ± â ÷ " chat completions â ÷ " metricsi¼%00 - **â|â¹ èf½âš"â¼°**i¼šè ÷ ââ|ç" Yç%00â¿jæð-âëðçç-æ³ • âëðK8s ç - %00âðšëç ÷ âYçYçYè - ÷ i¼Æçç-çç«âÆæ^ðéj¼ç" - **3 â¹ Python â • ¥ç" 'âÆ-ç»ðéâÆ**i¼šâ»žç š "ç" è,,šæœ-â°â" 'â¹³âð°â • ¥â... • çš,,âðÆæ • 'è - â¾,,